# Archisman Panigrahi

Graduate Student (Ph.D. Candidate) · Physics

*Massachusetts Institute of Technology, Cambridge, MA, USA*

+1 (857) 706-9484 | archi137@mit.edu | www.mit.edu/~archi137/ | Google Scholar


## Education

### Ph.D. in Physics (ongoing)
*Cambridge, MA, USA*

MASSACHUSETTS INSTITUTE OF TECHNOLOGY — *August 2022 - Ongoing*

- C.G.P.A - $5.0/5.0$

Supervisor: Prof. Leonid Levitov

### Master of Science in Physics
*Bangalore, India*

INDIAN INSTITUTE OF SCIENCE — *Aug. 2021 - Jun. 2022*

- C.G.P.A - $9.8/10.0$

### Bachelor of Science (Research) in Physics
*Bangalore, India*

INDIAN INSTITUTE OF SCIENCE — *Aug. 2017 - Jun. 2021*

- C.G.P.A - $9.8/10.0$

## Research Articles

LISTED IN THE CHRONOLOGICAL ORDER OF THEIR APPEARANCE ON ARXIV.

- **A. Panigrahi**, V. Poliakov, L. Levitov; *Particle-Hole Ghost Interference in Superconductors* arXiv:2606.06437
- Y. Liao, A. Grankin, **A. Panigrahi**, V. Galitski, L. Levitov; *Tunable viscosity across the BCS-BEC crossover* arXiv:2604.10759
- **A. Panigrahi**, D. Kaplan; *Valley polarization of chiral excitonic bound states induced by band geometry* arXiv:2604.05222
- P. Zhu, **A. Panigrahi**, L. Levitov, N. Trivedi; *Strain Response as a Probe of Spinons in Quantum Spin Liquids* arXiv:2512.03137
- **A. Panigrahi**, B. Roy; *Projected branes as platforms for crystalline, superconducting, and higher-order topological phases* PRB **113**, 035301 (2026)
- **A. Panigrahi**, V. Poliakov, L. Levitov; *Tomographic imaging of superconducting order using particle-hole interference* (accepted in PNAS) arXiv:2503.16168
- **A. Panigrahi**, A. Kumar; *Non-Fermi Liquids from Subsystem Symmetry Breaking in van der Waals Multilayers* PRL **134**, 236502 (2025)
- **A. Panigrahi**, V. Poliakov, Z. Dong, L. Levitov; *Spin chirality and fermion stirring in topological bands* arxiv:2407.17433
- L. Holleis, T. Xie, S. Xu, H. Zhou, C. L. Patterson, **A. Panigrahi**, T. Taniguchi, K. Watanabe, L. S. Levitov, C. Jin, E. Berg, A. F. Young; *Fluctuating magnetism and Pomeranchuk effect in multilayer graphene* Nature **640**, 355–360 (2025)
- M. Masseroni, M. Gull, **A. Panigrahi**, N. Jacobsen, F. Fischer, C. Tong, J. D. Gerber, M. Niese, T. Taniguchi, K. Watanabe, L. Levitov, T. Ihn, K. Ensslin, H. Duprez; *Spin-orbit proximity in $MoS_2$/bilayer graphene heterostructures* Nat Commun **15**, 9251 (2024)
- **A. Panigrahi**, L. Levitov; *Signatures of electronic ordering in transport in graphene flat bands* PRB **110**, 035122 (2024)
- **A. Panigrahi**, V. Juričić, B. Roy; *Projected Topological Branes* Commun Phys **5**, 230 (2022)
- **A. Panigrahi**, S. Mukerjee; *Energy magnetization and transport in systems with a non-zero Berry curvature in a magnetic field* SciPost Phys. Core 6, 052 (2023)
- **A. Panigrahi**, R. Moessner, B. Roy; *Non-Hermitian dislocation modes: Stability and melting across exceptional points* PRB **106**, L041302 (2022)

## Research Experience

### Aspects of spin chirality and tunneling density of states in systems with broken time-reversal symmetry
*MIT, Cambridge, MA, USA*

WITH PROF. LEONID LEVITOV — *2024 — Present*

- Demonstrated that tunneling density of states near two impurities may be utilized to probe superconducting order parameter
- Demonstrated that spin chirality is spontaneously generated in time-reveral symmetry broken systems without any spin-orbit coupling
- Predicted that this effect can be utilized in detecting topological phase winding

### Non-Fermi liquids resulting from subsystem symmetry breaking
*MIT, Cambridge, MA, USA*

WITH AJESH KUMAR — *2024*

- Demonstrated that subsystem symmetry breaking in van der Waals heterostructures can give rise to an anisotropic non-Fermi liquid, with quasiparticle lifetime $\tau \sim \frac{1}{|\omega| \log |1/\omega|}$ and specific heat $C \sim T[\log(1/T)]^2$.

### Transport in ordered phases in graphene
*MIT, Cambridge, MA, USA*

WITH PROF. LEONID LEVITOV
*2023 — 2024*

- Predicted that momentum-polarized nematic phases in biased bilayer graphene can lead to resistance decreasing with rising temperature
- Demonstrated hysteresis-like switching behavior under the action of a strong electric field

### Many Body Localization (MBL) and thermalization of interacting quantum spin chain
*IISc, Bangalore, India (Master's thesis)*

WITH PROF. SUBROTO MUKERJEE
*September 2021 - April 2022*

- Studied how the Out-of-Time Ordered Correlator (OTOC) behaves for MBL and thermal systems
- Studied behavior of OTOC in MBL systems with random and incommensurate potential, with and without interaction

### Topological phases in projected lower dimensional branes
*MPIPKS, Dresden, Germany (remotely)*

JOINTLY WITH PROF. BITAN ROY AND PROF. VLADIMIR JURIČIĆ
*June 2021 - September 2021*

- Verified the existence of dislocation modes, Weyl points, and Landau levels in projected crystals and Fibonacci quasicrystals
- Proposed how this method can be utilized to study higher dimensional (>3D) topological phases within 3D systems

### Berry curvature effects on thermoelectric transport
*IISc, Bangalore, India (Bachelor's thesis)*

WITH PROF. SUBROTO MUKERJEE
*October 2020 - June 2021*

- Found a condition on the energy magnetization such that the Einstein relation holds for the transport energy current in these systems
- Analytically solved the Boltzmann transport equation (including Berry curvature effects) for two-dimensional systems

### Non-Hermitian Topological Insulators and Dislocations
*MPIPKS, Dresden, Germany (remotely)*

WITH PROF. BITAN ROY
*May 2020 - September 2020*

- Obtained phase diagrams for regimes where topological states get pinned at dislocation centers
- Proposed how dislocations can be used to probe topological phases in non-Hermitian systems, where the non-Hermitian skin effect masks the traditional bulk-boundary correspondence

## Research Interests

### Broadly interested in theoretical Condensed Matter Physics

- **Non-Fermi Liquids emerging due to subsystem symmetry breaking**
- **Spin chirality in systems with spontaneously broken time-reversal symmetry**
- **Electronic transport in two-dimensional systems and the effects of Berry curvature in transport**
- **Computational methods in quantum condensed matter physics**
- **Topological phases of matter and Quantum Phase transitions**

## Skills

| | |
|---|---|
| **Programming skills** | Julia, MATLAB/Octave, Mathematica, Python |
| **Advanced Physics Courses** | Strongly Correlated Systems, Advanced Statistical Physics, Quantum Field Theory I, General Relativity |
| **Languages** | Fluent in English, Bengali, Hindi |

## Talks

### Tunneling density of states in exotic superconductors and spatial patterns of particle-hole interference
*APS March Meeting, Anaheim, CA*

CLICK HERE TO DOWNLOAD THE PRESENTATION
*March 2025*

### Transport Signatures of Electronic Ordering in Graphene Flat Bands
*Indian Institute of Science, Bangalore, India*

CLICK HERE TO DOWNLOAD THE PRESENTATION
*January 2024*

### Topological phases in quasicrystals: A general principle of construction
*APS March Meeting (virtually)*

CLICK HERE TO DOWNLOAD THE PRESENTATION
*March 2022*

### Dislocation as a bulk probe of non-Hermitian topology
*MPIPKS, Dresden, Germany (remotely)*

CLICK HERE TO DOWNLOAD THE PRESENTATION
*July 6, 2021*

## Teaching Experience

### Theory of Solids II (8.512)
*MIT*

Teaching Assistant — *Spring 2026*

- Conducted recitation classes on various topics in condensed matter physics

### Physics II: Electricity and Magnetism (8.102)
*MIT*

Teaching Assistant — *Spring 2024*

- Taught students one-on-one in office hours and graded exams

## Academic Achievements

| | | |
|---|---|---|
| 2023 | Qualified among the top 16 participants in MIT Integration Bee | *MIT* |
| 2022 | **1st Rank in India** in CSIR-NET (JRF) 2021 in Physics, held in February 2022 due to COVID (score 186/200) | *India* |
| 2022 | **1st Rank in India** in Graduate Aptitute Test in Engineering (G.A.T.E.) in Physics | *India* |
| 2017-22 | CGPA **9.8/10** in B.S. (Research) and M.S., received **Prof. R. Srinivasan Medal** for **highest CGPA** in batch | *IISc, Bangalore* |
| 2017 | **1st rank (99.2 %) in Board** in Higher Secondary Examination, among about 0.7 million candidates | *West Bengal, India* |
| 2015 | **2nd rank (97.57 %) in Board** in Secondary Examination, among about 1 million candidates | *West Bengal, India* |

## References

- Prof. **Leonid Levitov**, Dept. of Physics, Massachusetts Institute of Technology, Cambridge, MA 02139, USA.
  Email Address - levitov@mit.edu
- Prof. **Subroto Mukerjee**, Dept. of Physics, Indian Institute of Science, Bangalore, India.
  Email Address - smukerjee@iisc.ac.in
- Prof. **Bitan Roy**, Dept. of Physics, Lehigh University, Bethlehem, PA 18015, USA.
  Email Address - bitan.roy@lehigh.edu